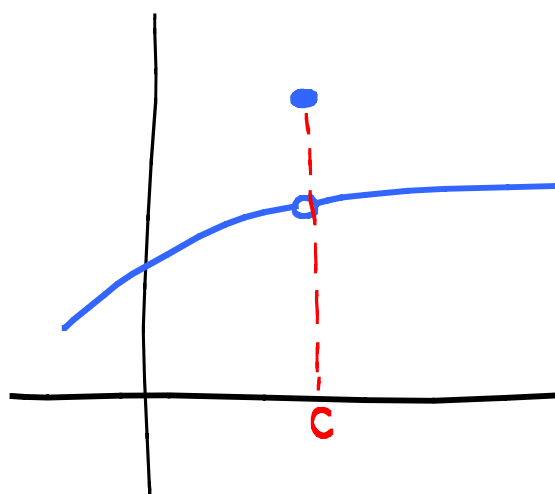
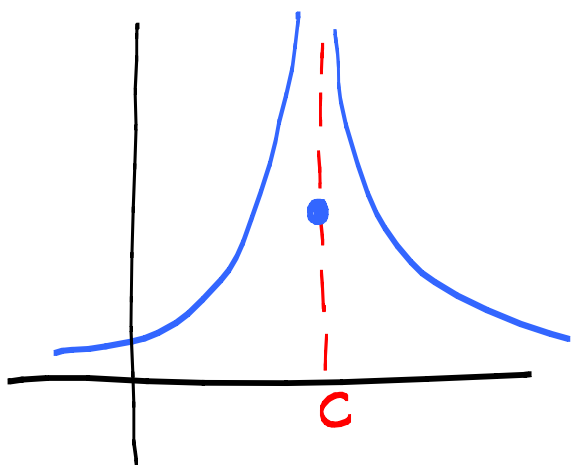
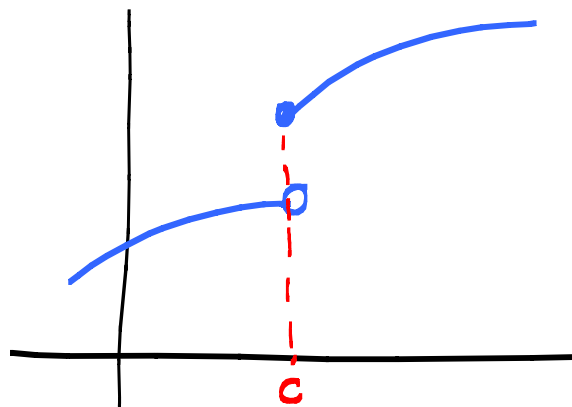
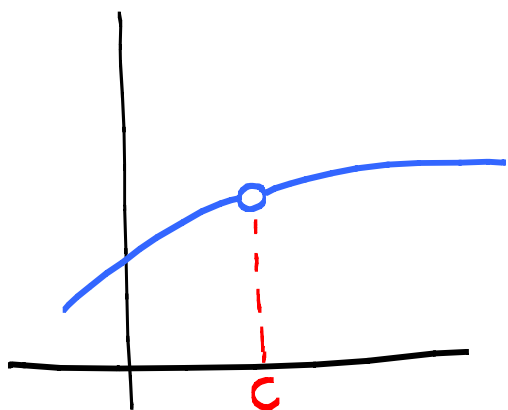


Unit 2.4 Continuity

pp 118-130

Note Title

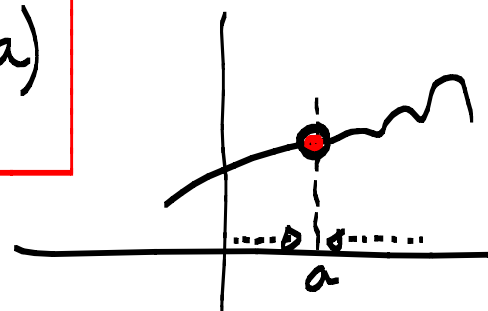
2014/02/27



* Definition 1: p 118

A function f is continuous at a number $x = a$ if

$$\lim_{x \rightarrow a} f(x) = f(a)$$



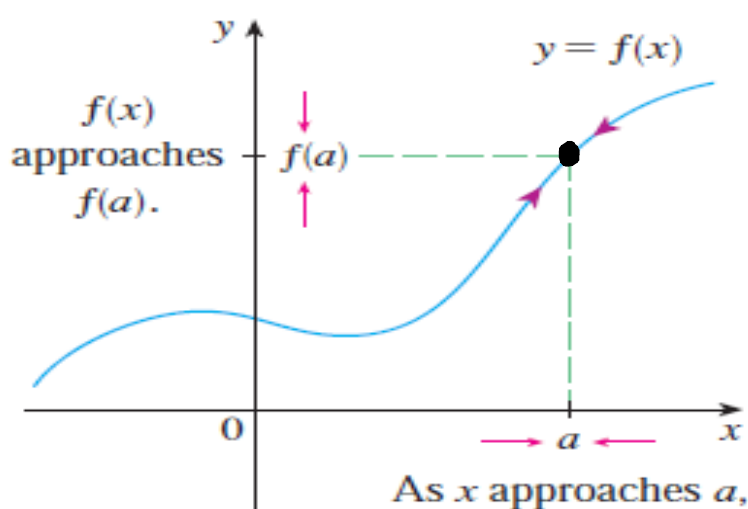
Test 3 things:

- (i) $f(a)$ must exist
- (ii) $\lim_{x \rightarrow a} f(x)$ must exist:

$$\left[\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L \right]$$

- (iii) $\lim_{x \rightarrow a} f(x) = f(a)$

If one of these fail then f is discontinuous in the point a .



Examples:

- ① Show that $f(x) = x^2 - 1$ is continuous at $x = 2$

(i) $f(2) = 3$ it exists.

$$(ii) \lim_{x \rightarrow 2^-} x^2 - 1 = \lim_{x \rightarrow 2^+} x^2 - 1 = 3$$

$$\therefore \lim_{x \rightarrow 2} x^2 - 1 = 3 \Rightarrow \text{it exists.}$$

$$(iii) f(2) = \lim_{x \rightarrow 2} (x^2 - 1)$$

$\Rightarrow f$ is continuous at $x=2$

$$(2) f(x) = \begin{cases} \frac{x^2-9}{x-3} & ; x \neq 3 \\ 6 & ; x = 3 \end{cases}$$

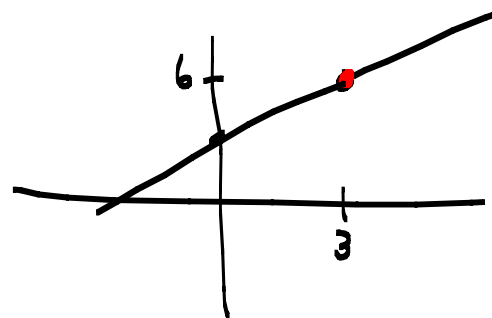
Is f continuous at $x=3$?

$$(i) f(3) = 6$$

$$(ii) \lim_{x \rightarrow 3} \frac{(x-3)(x+3)}{\cancel{x-3}} = 6$$

$$(iii) f(3) = \lim_{x \rightarrow 3} f(x) = 6$$

$\Rightarrow f$ continuous at $x=3$.

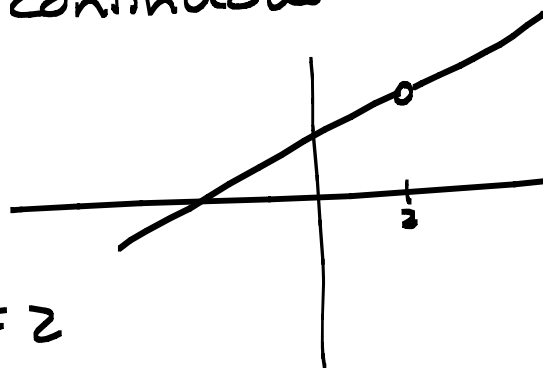


$$(3) f(x) = \frac{x^2-4}{x-2} \text{ is } f \text{ continuous at}$$

$$x=2.$$

$$(i) f(2) \text{ DNE}$$

$\Rightarrow f$ is discont. in $x=2$



$$f(x) = \frac{(x-2)(x+2)}{x-2} = x+2, x \neq 2$$

$$\textcircled{4} \quad f(x) = \begin{cases} \frac{3x}{x^2+x} & \text{if } x > 0 \leftarrow 0^+ \mathbb{R} \\ \textcircled{3} & \text{if } x = 0 \leftarrow \\ \frac{\sin 3x}{x} & \text{if } x < 0 \leftarrow 0^- \end{cases}$$

Is f continuous at $x = 0$?

$$(i) \quad f(0) = 3$$

$$\begin{aligned} (ii) \quad \lim_{x \rightarrow 0^-} \left(\frac{\sin 3x}{x} \right) \left(\frac{0}{0} \right) &= \lim_{x \rightarrow 0^-} \frac{\textcircled{3} \sin 3x}{\textcircled{3} x} \\ &= 3 \end{aligned} \quad \begin{aligned} \lim_{x \rightarrow 0^+} \frac{3x}{x^2+x} &\left(\frac{0}{0} \right) \\ &= \lim_{x \rightarrow 0^+} \frac{x(3)}{x(\textcircled{x+1})} \\ &= 3 \end{aligned}$$

$$\lim_{x \rightarrow 0} f(x) = 3$$

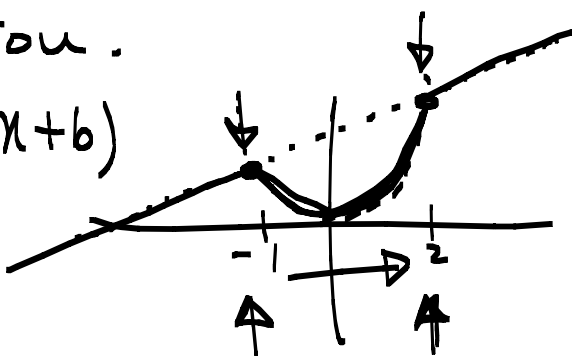
$$(iii) \quad f(0) = \lim_{x \rightarrow 0} f(x)$$

$\Rightarrow f$ continuous at $x = 0$

$$\textcircled{5} \quad f(x) = \begin{cases} x^2 & \text{if } -1 \leq x \leq 2 \\ ax+b & \text{if } x < -1 \text{ or } x > 2 \end{cases}$$

Determine a and b if f is continuous on $\mathbb{R}$. Sketch the function.

$$\begin{aligned} (i) \quad f(-1) &= 1 \quad \lim_{x \rightarrow -1^-} f = \lim_{x \rightarrow -1^-} (ax+b) \\ &= -a+b \end{aligned}$$



$$\lim_{x \rightarrow -1^+} f = \lim_{x \rightarrow -1^+} x^2 = 1$$

$$\Rightarrow -a + b = 1 \quad \text{————— (1)}$$

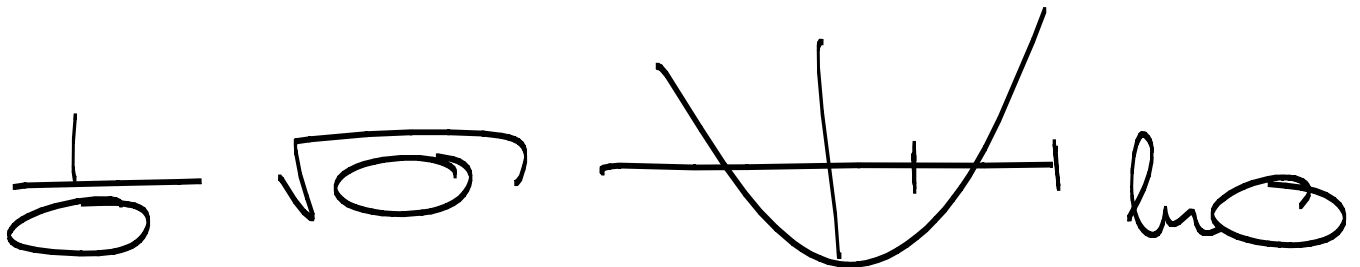
$$(ii) \quad f(2) = 4 \quad \lim_{x \rightarrow 2^-} f = 4 \quad \lim_{x \rightarrow 2^+} f = 2a + b$$

$$\Rightarrow 2a + b = 4 \quad \text{————— (2)}$$

$$\textcircled{1} - \textcircled{2} \quad 3a = 3 \quad \Rightarrow a = 1$$

$$\Rightarrow b = 2$$

$$f(x) = x + 2$$



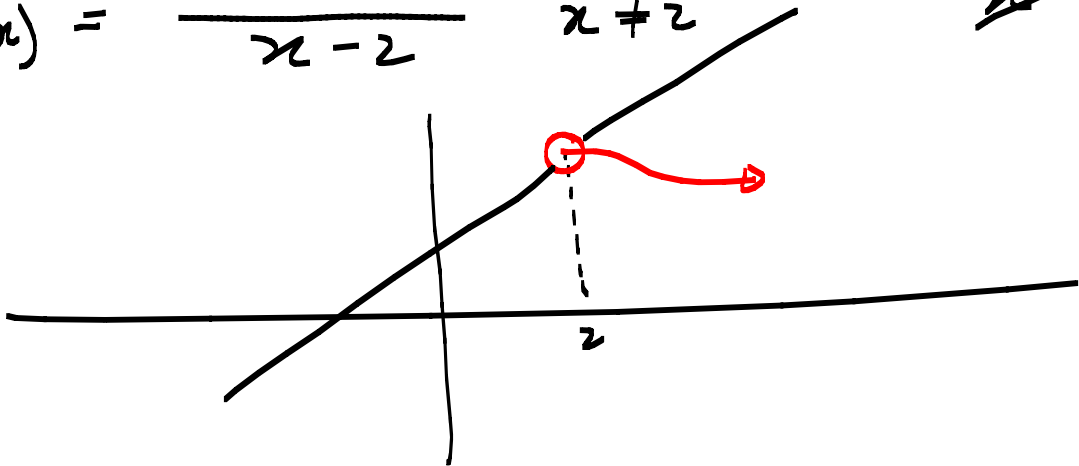
Definition

If f is not continuous at a , then we say f is discontinuous at $x = a$.

Removable discontinuity

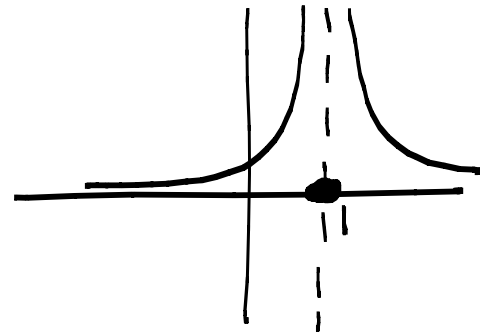
$$f(x) = \frac{x^2 - x - 2}{x - 2} \quad x \neq 2$$

$$\frac{(x+1)\cancel{(x-2)}}{\cancel{x-2}}$$



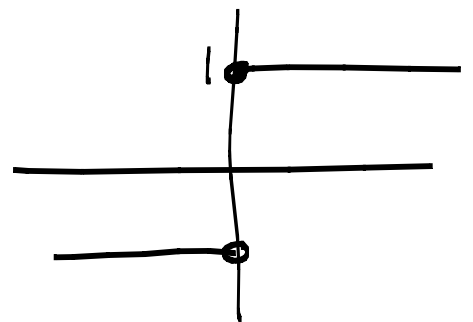
Infinite discontinuity

$$f(x) = \begin{cases} \frac{1}{(x-1)^2} & \text{if } x \neq 1 \\ \underline{\underline{0}} & \text{if } x = 1 \end{cases}$$



Jump discontinuity.

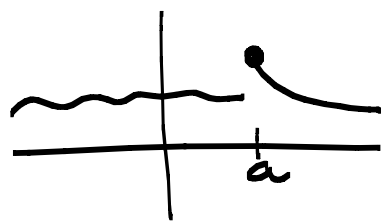
$$f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$$



Definition 2 p120

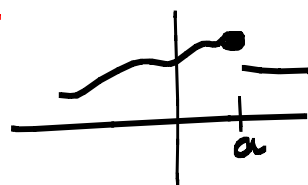
A function f is continuous from the right at a number a if:

$$\lim_{x \rightarrow a^+} f(x) = f(a)$$



and continuous from the left at a if:

$$\lim_{x \rightarrow a^-} f(x) = f(a)$$



Example: $f(x) = \begin{cases} 1+x & \text{if } x \leq 1 \\ 2+x & \text{if } x > 1 \end{cases} *$

(i) $f(1) = 2$

(ii) $\lim_{x \rightarrow 1^-} (1+x)$
 $= 2$

$$\lim_{x \rightarrow 1^+} (2+x) = 3$$

$$\lim_{x \rightarrow 1} f(x) \text{ DNE.}$$

(iii) $\lim_{x \rightarrow 1^-} f(x) = f(1) \Rightarrow f$ is continuous from the left!

Definition 3: p 120 SELF STUDY.

Continuity on intervals.

Theorem 4: p 121 (leave proof)

If f and g are continuous at a , then the following functions are also continuous.

1. $f+g$ 2. $f-g$ 3. $c \cdot f$ $\hookrightarrow$ constant

4. fg 5. $\frac{f}{g}$ if $g(a) \neq 0$

Theorem 5 and 7: p 122 & 123

These functions are continuous on their domains:

polynomials, rational functions,
root functions, trig functions,
inverse trig functions,
exponential and logarithmic functions.

Examples

① Where is $f(x) = \frac{\ln x + \tan^{-1} x}{x^2 - 1}$

continuous?

$\lim_{x \rightarrow 0} \frac{1}{x}$

(i) $\ln x$ — log. function, Conti. on domain.
Conti for $x > 0$.

(ii) $\tan^{-1} x$ — conti on $\mathbb{R}$.

(iii) $x^2 - 1$ — polynomial is conti on $\mathbb{R}$.
but $x^2 - 1 \neq 0 \Rightarrow x = \pm 1$

f continuous on $(0, \infty) \setminus \{1\}$ or
 $(0, 1) \cup (1, \infty)$

Theorem 8 p 124

If $\lim_{x \rightarrow a} g(x) = b$ and f continuous
at b then

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right) = f(b)$$

Example:

$$\lim_{x \rightarrow 1} \arcsin\left(\frac{1 - \sqrt{x}}{1 - x}\right)$$

$$\begin{aligned}
&= \arcsin \left(\lim_{x \rightarrow 1} \frac{1 - \sqrt{x}}{1 - x} \right) \quad \left(\frac{0}{0} \right) \\
&= \arcsin \left(\lim_{x \rightarrow 1} \frac{1 - \sqrt{x}}{(1 - \sqrt{x})(1 + \sqrt{x})} \right) \\
&= \arcsin \left(\frac{1}{2} \right) = \frac{\pi}{6}
\end{aligned}$$

Theorem 9 p 125

g is continuous at $x=a$ and f continuous at $g(a)$ then $f \circ g$ is continuous at a .
 $f(g(x))$

Example: Where is $h(x) = \sin(e^x)$ continuous?

$$h(x) = \sin(e^x) \quad \text{so} \quad \begin{aligned} g(x) &= e^x \\ f(x) &= \sin x \end{aligned}$$

$$\text{then } h(x) = f(g(x)) = (f \circ g)(x)$$

- g continuous on $\mathbb{R}$, f continuous on $\mathbb{R}$
 so $f \circ g$ is continuous on $\mathbb{R}$.